Mukul Rayana

AI Engineer | Forward Deployed Engineer

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Summary

AI engineer who builds agent and retrieval systems that real people use. Won 1st place at the UMD Smith Agentic AI Challenge, runs a public RAG service with live users, and cut a production model's latency 3x as the only ML engineer on an XR team. Good at turning a vague client ask into a shipped fix.

Publication

Voice-Driven Panoramic Imagery: Real-Time Generative AI for Immersive Experiences | IEEE Xplore IDCIoT 2024

- Co-author. Built speech-to-360-degree image generation with Whisper and Stable Diffusion at **2.3x** throughput.

Experience

BeeBox Studios Pvt Ltd | IIT Madras Research Park Apr 2023 - Apr 2024
AI Research Intern Chennai, India

- Turned a loose "make it faster" request into a 3x latency cut for field technicians on XR headsets, from **185 ms to 62 ms** P95, with ONNX export and INT8 quantization of a BERT question-answering model (**83.2% F1**).
- Delivered client-requested multi-peer WebRTC a day ahead of a four-day deadline after taking over an unfamiliar JavaScript and Python module, then tested it end to end on a live call.
- Connected Python inference to the Unity/C# client over TCP as the only ML engineer, and deployed to AWS EC2 with Docker and GitHub Actions CI/CD, supporting **six production releases**.
- Proposed a local model over a paid LLM API to cut token cost and keep documents in-house, which the team later adopted.

Education

University of Maryland, College Park | *M.S., Applied Machine Learning* Aug 2024 - May 2026
SRM Easwari Engineering College | *B.Tech., Artificial Intelligence & Data Science* Dec 2020 - Apr 2024
Certifications: Data Structures & Algorithms Specialization, UC San Diego; IBM Data Science Professional Certificate.

Projects

PharmaChain: Cold-Chain Shipment Incident Response | Live Demo | GitHub Apr 2026

- Won **1st place ($4,000)** in a team of five. Eight agents take a shipment alert to an approved action plan in **12 to 15 seconds**, versus the usual 6 to 8 hours.
- Built the Route and Triage agents, their REST API and an LLM router that fails over across four providers to a rule-based mode, and designed the insurance step to always require human approval.

Irminsul: Public Q&A Service with Cited Answers | Live Demo | GitHub Nov 2025 - Present

- Built and run a service used by **15 Genshin Impact players**: **210 requests/s at 0.45 s p95** with zero errors in a 50-user load test, right source in the top 8 for **94.6%** of test questions, and **113 of 118** live tests passing, including all 19 safety probes.

Promise Ledger: Checking Affordable-Housing Commitments | Live Demo | Video | GitHub Nov 2026

- Led design and built with AI coding tools, in a team of two for GDG AI for Good 2026, a tool that checks the affordable-housing commitments developers make in exchange for tax breaks, tracing each to the exact page of the public record.
- Code keeps a quote only on a word-for-word page match: **96.3% precision** and **98.1% recall** on a 53-claim development set, about **$0.002 per PDF** on Gemini, **258 tests**, on Google Cloud Run.

RECON: Stale Evidence Detection for Research Search | Live Demo | GitHub Oct 2025 - Jan 2026

- Built a four-agent LangGraph pipeline that catches **44%** of stale claims on a 130-question benchmark, versus **0%** for single-pass RAG, with 2% false positives.

Technical Skills

Languages & Tools: Python, C#, SQL; data structures and algorithms; AI coding assistants
AI/ML: Generative AI, RAG, agentic systems, LangGraph, NLP, QLoRA, PyTorch, TensorFlow, scikit-learn
Integration & Deployment: REST APIs, FastAPI, Docker, GitHub Actions CI/CD, AWS (EC2, S3), Google Cloud Run
Data & Search: PostgreSQL, MongoDB, SQLite, pgvector, Pinecone, FAISS